Post-groups, post-groupoids and the Yang-Baxter equation

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Post-Lie algebras

Definition (Vallette)

A **post-Lie algebra** $(\mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}}, \triangleright)$ consists of a Lie algebra $(\mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}})$ and a binary product $\triangleright : \mathfrak{g} \otimes \mathfrak{g} \to \mathfrak{g}$ such that

$$\begin{aligned} x \triangleright [y,z]_{\mathfrak{g}} &= [x \triangleright y, z]_{\mathfrak{g}} + [y, x \triangleright z]_{\mathfrak{g}}, \\ [x,y]_{\mathfrak{g}} \triangleright z &= a_{\triangleright}(x,y,z) - a_{\triangleright}(y,x,z), \end{aligned}$$

where $a_{\triangleright}(x,y,z) = x \triangleright (y \triangleright z) - (x \triangleright y) \triangleright z$.



B. Vallette, Homology of generalized partition posets, *J. Pure Appl. Algebra* **208** (2007), 699-725.

$$\begin{aligned} L^{\triangleright}_x [y,z]_{\mathfrak{g}} &= [L^{\triangleright}_x y, z]_{\mathfrak{g}} + [y, L^{\triangleright}_x z]_{\mathfrak{g}}, \\ ([x,y]_{\mathfrak{g}} + x \triangleright y - y \triangleright x) \triangleright z &= x \triangleright (y \triangleright z) - y \triangleright (x \triangleright z). \end{aligned}$$

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Post-Lie algebras

Proposition

A post-Lie algebra $(\mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}}, \triangleright)$ *gives rise to a new Lie algebra* $(\mathfrak{g}, [\cdot,\cdot]_{\triangleright})$, *called the* **sub-adjacent Lie algebra** *and denoted by* $\mathfrak{g}_{\triangleright}$, *where the Lie bracket* $[\cdot,\cdot]_{\triangleright}$ *is defined by*

$$[x,y]_{\triangleright} = [x,y]_{\mathfrak{g}} + x \triangleright y - y \triangleright x, \ \forall x, y \in \mathfrak{g}.$$

Moreover, $L^{\triangleright} : (\mathfrak{g}, [\cdot,\cdot]_{\triangleright}) \to \mathrm{Der}(\mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}})$ *is an action of the sub-adjacent Lie algebra* $\mathfrak{g}_{\triangleright}$ *on the original Lie algebra* $(\mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}})$.

In a post-Lie algebra $(\mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}}, \triangleright)$, if the Lie bracket $[\cdot,\cdot]_{\mathfrak{g}}$ is trivial, then we obtain a pre-Lie algebra, namely a vector space $\mathfrak{g}$ with a multiplication $\triangleright$ satisfying

$$a_{\triangleright}(x,y,z) - a_{\triangleright}(y,x,z) = 0.$$

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$$a_{\triangleright}(x,y,z) - a_{\triangleright}(y,x,z) = 0.$$

Post-Lie algebras

Example

Let $(\mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}})$ be a Lie algebra, $(A, \cdot)$ a commutative associative algebra and $\rho : \mathfrak{g} \to \mathrm{Der}(A)$ a Lie algebra homomorphism. Then there is a post-Lie algebra structure on $A \otimes \mathfrak{g}$, which is given by

$$\begin{aligned} [a \otimes x, b \otimes y] &= ab \otimes [x,y]_{\mathfrak{g}}, \\ (a \otimes x) \rhd (b \otimes y) &= a \cdot \rho(x)(b) \otimes y, \end{aligned}$$

for all $a, b \in A,\ x, y \in \mathfrak{g}$.

Examples

Example

Let $(V, \triangleright)$ be a magma algebra. Extend the magma operation $\triangleright : V \otimes V \to V$ to the free Lie algebra $(Lie(V), [\cdot, \cdot])$. Then $(Lie(V), [\cdot, \cdot], \triangleright)$ is a post-Lie algebra.

Examples

Example (Free post-Lie algebra)

Let $\mathcal{O}$ be the set of isomorphism classes of planar rooted trees:

$$\mathcal{PT} = \Big\{ \bullet, \bullet^{\bullet}, \bullet^{\bullet\bullet}, \bullet^{\bullet^{\bullet}}, \bullet^{\bullet\bullet\bullet}, \bullet^{\bullet^{\bullet\bullet}}, \bullet^{\bullet\bullet^{\bullet}}, \bullet^{\bullet^{\bullet^{\bullet}}}, \bullet^{\bullet\bullet\bullet\bullet}, \bullet^{\bullet^{\bullet}\bullet^{\bullet}}, \bullet^{\bullet^{\bullet\bullet\bullet}}, \bullet^{\bullet\bullet^{\bullet\bullet}}, \bullet^{\bullet^{\bullet}\bullet\bullet^{\bullet}}, \bullet^{\bullet^{\bullet^{\bullet\bullet}}}, \bullet^{\bullet\bullet^{\bullet^{\bullet}}}, \bullet^{\bullet^{\bullet^{\bullet}}\bullet^{\bullet}}, \bullet^{\bullet^{\bullet^{\bullet^{\bullet}}}}, \ldots \Big\}.$$

Let $\mathbf{k}\{\mathcal{O}\}$ be the free $\mathbf{k}$-vector space generated by $\mathcal{O}$. The **left grafting operator** $\triangleright : \mathbf{k}\{\mathcal{O}\} \otimes \mathbf{k}\{\mathcal{O}\} \to \mathbf{k}\{\mathcal{O}\}$ is defined by

$$\tau \triangleright \omega = \sum_{s \in \mathrm{Nodes}(\omega)} \tau \circ_s \omega, \ \forall \tau, \omega \in \mathcal{O},$$

where $\tau \circ_s \omega$ is the planar rooted tree resulting from attaching the root of τ to the node s of the tree ω from the left. Consider the free Lie algebra $Lie(\mathbf{k}\{\mathcal{O}\})$, and extend the left grafting operator $\triangleright$ on $\mathbf{k}\{\mathcal{O}\}$ to the free Lie algebra $Lie(\mathbf{k}\{\mathcal{O}\})$. Then $(Lie(\mathbf{k}\{\mathcal{O}\}), [\cdot,\cdot], \triangleright)$ is a post-Lie algebra, which is the free post-Lie algebra generated by one generator $\{\bullet\}$.

Examples

Example (Free pre-Lie algebra)

Let $\mathcal{T}$ be the set of isomorphism classes of rooted trees, which is denoted by

$$\mathcal{T} = \Big\{ \bullet, \bullet\!\!\bullet, \vee, \bullet\!\!\bullet\!\!\bullet, \bullet\!\!\vee, \vee\bullet, \bullet\!\!\bullet\!\!\bullet\!\!\bullet, \bullet\vee, \vee\!\!\bullet, \bullet\!\!\vee\!\!\bullet, \vee\!\!\vee, \bullet\vee\bullet, \vee\bullet\!\!\bullet, \bullet\!\!\bullet\vee, \bullet\!\!\bullet\!\!\bullet\!\!\bullet\!\!\bullet, \ldots \Big\}.$$

Let $\mathbf{k}\{\mathcal{T}\}$ be the free $\mathbf{k}$-vector space generated by $\mathcal{T}$. The **grafting operator** $\curvearrowright\colon \mathbf{k}\{\mathcal{T}\} \times \mathbf{k}\{\mathcal{T}\} \to \mathbf{k}\{\mathcal{T}\}$ is defined by

$$\tau \curvearrowright \omega = \sum_{s \in \mathrm{Nodes}(\omega)} \tau \circ_s \omega, \quad \forall \tau, \omega \in \mathcal{T},$$

where $\tau \circ_s \omega$ is the rooted tree resulting from attaching the root of τ to the node s of the rooted tree ω. Chapoton and Livernet have shown that $(\mathbf{k}\{\mathcal{T}\}, \curvearrowright)$ is the **free pre-Lie algebra** generated by one generator $\{\bullet\}$.

Examples

Example

There is a natural pre-Lie algebra structure on the vector space $\mathfrak{X}(\mathbb{R}^n)$ of smooth vector fields, which is given by

$$\sum_{1\leq i\leq n} X_i\partial_i \rhd \sum_{1\leq j\leq n} Y_j\partial_j = \sum_{1\leq i,j\leq n} X_i(\partial_i Y_j)\partial_j,$$

for all $X_i, Y_j \in C^\infty(\mathbb{R}^n)$.

Butcher series

We consider systems of differential equations

$$\begin{cases} \frac{\mathrm{d}y_1}{\mathrm{d}x} &= f_1(y_1, y_2, \cdots, y_n), \\ \frac{\mathrm{d}y_2}{\mathrm{d}x} &= f_2(y_1, y_2, \cdots, y_n), \\ &\vdots \\ \frac{\mathrm{d}y_n}{\mathrm{d}x} &= f_n(y_1, y_2, \cdots, y_n). \end{cases}$$

Use the vector notation as following:

$$y = (y_1, y_2, \cdots, y_n)^T, \quad f(y) = (f_1(y), f_2(y), \cdots, f_n(y))^T,$$

we can rewrite the above systems of differential equations to $\frac{\mathrm{d}y}{\mathrm{d}x} = f(y)$ with the initial value

$$y(x_0) = y_0 = (y_1(x_0), y_2(x_0), \cdots, y_n(x_0))^T.$$

Butcher series

It is really surprise that one should use the free pre-Lie algebra in the formal Taylor expansion of the solution at $x_0 + h$.

Any map $f : \mathbb{R}^n \to \mathbb{R}^n$ gives a smooth vector field

$$f(y) = \sum_{1 \leq i \leq n} f_i(y)\partial_i \in \mathfrak{X}(\mathbb{R}^n).$$

Moreover, there is a unique pre-Lie algebra homomorphism F from $\mathbb{R}\{\mathcal{T}\}$ to $\mathfrak{X}(\mathbb{R}^n)$ such that

$$F(\bullet) = f.$$

Applications in Butcher series

The formal Taylor expansion of the solution at $x_0 + h$ is

$$y(x_0 + h) = y_0 + \sum_{\tau \in \mathcal{T}} \frac{h^{|\tau|}}{\sigma(\tau)\tau!} F(\tau)(y_0).$$

Set $\mathcal{T}^+ = \mathcal{T} \cup \{\emptyset\}$ and denote by $\mathcal{B}_{\mathbb{R}}$ the set of all maps $\{a : \mathcal{T}^+ \to \mathbb{R} | a(\emptyset) = 1\}$. The formal series defined by

$$B(a, h, f)(y) = a(\emptyset)y + \sum_{\tau \in \mathcal{T}} \frac{h^{|\tau|} a(\tau)}{\sigma(\tau)} F(\tau)(y)$$

is called the **Butcher series**.

 J. C. Butcher, An algebraic theory of integration methods, *Math. Comp.* **26** (1972), 79-106.

 H. Z. Munthe-Kaas and A. Lundervold, On post-Lie algebras, Lie-Butcher series and moving frames, *Found. Comput. Math.* **13** (2013), 583-613.

Applications in regularity structures

There is a meta theorem in the regularity structures.

Theorem (Bruned-Katsetsiadis)

The Hopf algebra in Martin Hairer's regularity structures encoding the combinatorics of recentering is obtained as the universal envelope of a suitable post-Lie algebra.


Y. Bruned and F. Katsetsiadis, Post-Lie algebras in regularity structures. *Forum Math. Sigma* **11** (2023), Paper No. e98.


M. Hairer, A theory of regularity structures. *Invent. Math.* **198** (2014), 269-504.


Y. Bruned, M. Hairer and L. Zambotti, Algebraic renormalisation of regularity structures. *Invent. Math.* **215** (2019), 1039-1156.

Applications in Rota-Baxter algebras

Rota-Baxter operators $\rightsquigarrow$ splitting of algebras

Proposition

Let $B : \mathfrak{g} \to \mathfrak{g}$ be a Rota-Baxter operator on a Lie algebra $\mathfrak{g}$, i.e. the following equality holds:

$$[B(x), B(y)]_{\mathfrak{g}} = B([B(x), y]_{\mathfrak{g}} + [x, B(y)]_{\mathfrak{g}} + [x, y]_{\mathfrak{g}}).$$

Define a multiplication $\triangleright_{\mathfrak{B}}$ on $\mathfrak{g}$ by

$$x \triangleright_B y = [B(x), y]_{\mathfrak{g}}, \quad \forall x, y \in \mathfrak{g}.$$

Then $(\mathfrak{g}, [\cdot, \cdot]_{\mathfrak{g}}, \triangleright_B)$ is a post-Lie algebra.

C. Bai, L. Guo and X. Ni, Nonabelian generalized Lax pairs, the classical Yang-Baxter equation and PostLie algebras, *Comm. Math. Phys.* 297 (2010), 553-596.

Applications in braces

A brace $(G, \circ, \cdot)$ consists of an abelian group $(G, \cdot)$ and a group $(G, \circ)$ such that

$$a \circ (b \cdot c) = (a \circ b) \cdot a^{\cdot -1} \cdot (a \circ c), \quad \forall a, b, c \in G.$$

Smoktunowicz proposed the following questions:

Question. Is there a passage from all left nilpotent braces of cardinality p^n, with $n + 1 < p$, to left nilpotent pre-Lie rings?



A. Smoktunowicz, On the passage from finite braces to pre-Lie rings. *Adv. Math.* **409** (2022), 108683.

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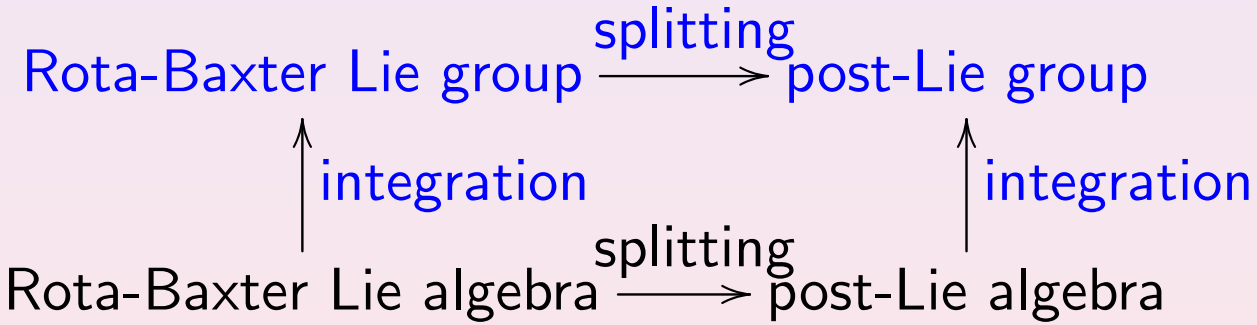
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Questions:

- What is the integration of post-Lie algebras, and Rota-Baxter Lie algebras?

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Definition (Bai-Guo-S.-Tang)

A **post-group** is a group $(G, \cdot)$ equipped with another binary operation $\triangleright$ on G such that

1. for all $a \in G$, the left multiplication

$$L_a^{\triangleright} : G \to G, \quad L_a^{\triangleright} b = a \triangleright b, \quad \forall b \in G,$$

is an automorphism of the group $(G, \cdot)$, that is,

$$a \triangleright (b \cdot c) = (a \triangleright b) \cdot (a \triangleright c), \quad \forall a, b, c \in G;$$

2. the following "weighted" associativity for $\triangleright$ holds:

$$a \triangleright (b \triangleright c) = \big(a \cdot (a \triangleright b)\big) \triangleright c, \quad \forall a, b, c \in G.$$

Post-groups

Theorem (Bai-Guo-S.-Tang)

Let $(G, \cdot, \triangleright)$ be a post-group. Define $\circ : G \times G \to G$ by

$$a \circ b \quad = \quad a \cdot (a \triangleright b), \quad \forall a, b \in G.$$

Then $(G, \circ)$ is a group with e being the unit, and the inverse map $\dagger : G \to G$ given by

$$a^{\dagger} \quad := \quad (L_a^{\triangleright})^{-1}(a^{-1}).$$

Moreover, $L^{\triangleright} : G \to \mathrm{Aut}(G)$ is an action of the group $(G, \circ)$ on the group $(G, \cdot)$.

The group $G_{\triangleright} := (G, \circ)$ is called the **subadjacent group** of the post-group $(G, \cdot, \triangleright)$.

Let e the identity of the Lie group G. Let $\mathfrak{g} = T_eG$ be the Lie algebra of G.

Define $\triangleright : \mathfrak{g} \otimes \mathfrak{g} \to \mathfrak{g}$ by

$$x \triangleright y = L^{\triangleright}_{*e}(x)(y) = \frac{d}{dt}\Big|_{t=0} L^{\triangleright}_{\exp(tx)} y = \frac{d}{dt}\Big|_{t=0} \frac{d}{ds}\Big|_{s=0} L^{\triangleright}_{\exp(tx)} \exp(sy).$$

Theorem (Bai-Guo-S.-Tang)

Let $(G, \cdot, \triangleright)$ be a post-Lie group. Then $(\mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}}, \triangleright)$ is a post-Lie algebra.

Let

$$\exp^{(\cdot)} : \mathfrak{g} \longrightarrow G$$

be the exponential map.

$$
\begin{aligned}
& x \triangleright (y \triangleright z) - y \triangleright (x \triangleright z) \\
= \; & \frac{d}{dt}\Big|_{t=0} \frac{d}{ds}\Big|_{s=0} \frac{d}{dr}\Big|_{r=0} \left(L^{\triangleright}_{\exp(tx)} L^{\triangleright}_{\exp(sy)} \exp(rz) - L^{\triangleright}_{\exp(sy)} L^{\triangleright}_{\exp(tx)} \exp(rz) \right) \\
= \; & \frac{d}{dt}\Big|_{t=0} \frac{d}{ds}\Big|_{s=0} \frac{d}{dr}\Big|_{r=0} \left(L^{\triangleright}_{\exp(tx)\cdot(\exp(tx)\triangleright\exp(sy))} \exp(rz) - L^{\triangleright}_{\exp(sy)\cdot(\exp(sy)\triangleright\exp(tx))} \exp(rz) \right) \\
= \; & \frac{d}{dt}\Big|_{t=0} \frac{d}{ds}\Big|_{s=0} \frac{d}{dr}\Big|_{r=0} L^{\triangleright}_{\exp(tx)\cdot\exp(sy)} \exp(rz) + \frac{d}{dt}\Big|_{t=0} \frac{d}{ds}\Big|_{s=0} \frac{d}{dr}\Big|_{r=0} L^{\triangleright}_{\exp(tx)\triangleright\exp(sy)} \exp(rz) \\
& - \frac{d}{dt}\Big|_{t=0} \frac{d}{ds}\Big|_{s=0} \frac{d}{dr}\Big|_{r=0} L^{\triangleright}_{\exp(sy)\cdot\exp(tx)} \exp(rz) - \frac{d}{dt}\Big|_{t=0} \frac{d}{ds}\Big|_{s=0} \frac{d}{dr}\Big|_{r=0} L^{\triangleright}_{\exp(sy)\triangleright\exp(tx)} \exp(rz) \\
= \; & \frac{d}{dt}\Big|_{t=0} \frac{d}{ds}\Big|_{s=0} \frac{d}{dr}\Big|_{r=0} L^{\triangleright}_{\exp(tx+sy+\frac{1}{2}ts[x,y]_{\mathfrak{g}}+\cdots)} \exp(rz) + (x \triangleright y) \triangleright z \\
& - \frac{d}{dt}\Big|_{t=0} \frac{d}{ds}\Big|_{s=0} \frac{d}{dr}\Big|_{r=0} L^{\triangleright}_{\exp(sy+tx+\frac{1}{2}ts[y,x]_{\mathfrak{g}}+\cdots)} \exp(rz) - (y \triangleright x) \triangleright z \\
= \; & \frac{1}{2}[x,y]_{\mathfrak{g}} \triangleright z + (x \triangleright y) \triangleright z - \frac{1}{2}[y,x]_{\mathfrak{g}} \triangleright z - (y \triangleright x) \triangleright z \\
= \; & [x,y]_{\triangleright} \triangleright z,
\end{aligned}
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Rota-Baxter Lie groups

Definition (Guo-Lang-S.)

A Rota-Baxter operator of weight 1 on a Lie group G is a smooth map $\mathfrak{B} : G \to G$ such that

$$\mathfrak{B}(g)\mathfrak{B}(h) = \mathfrak{B}(g\mathrm{Ad}_{\mathfrak{B}(g)}h), \qquad g, h \in G.$$

Theorem (Guo-Lang-S.)

If $(G, \mathfrak{B})$ is a Rota-Baxter Lie group, then $(\mathfrak{g}, B = \mathfrak{B}_{*e})$ is a Rota-Baxter Lie algebra of weight 1.

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Useful formulas

The relation between the Lie bracket $[\cdot,\cdot]_{\mathfrak{g}}$ and the Lie group multiplication is given by the following important formula:

$$[u,v]_{\mathfrak{g}} = \frac{d^2}{dtds}\Big|_{t,s=0} \exp^{tu}\exp^{sv}\exp^{-tu}, \quad \forall\ u,v \in \mathfrak{g}.$$

Since $B = \mathfrak{B}_{*e}$ is the tangent map of $\mathfrak{B}$ at e, we have the following relation for sufficiently small t:

$$\frac{d}{dt}\Big|_{t=0} \mathfrak{B}(\exp^{tu}) = \frac{d}{dt}\Big|_{t=0} \exp^{tB(u)} = B(u), \quad \forall\ u \in \mathfrak{g}.$$

Proof.

$$
\begin{aligned}
& [B(u), B(v)] \\
= \quad & \frac{d^2}{dtds}\Big|_{t,s=0} \exp^{tB(u)} \exp^{sB(v)} \exp^{-tB(u)} \\
= \quad & \frac{d^2}{dtds}\Big|_{t,s=0} \mathfrak{B}(\exp^{tu})\mathfrak{B}(\exp^{sv})\mathfrak{B}(\exp^{-tu}) \\
= \quad & \frac{d^2}{dtds}\Big|_{t,s=0} \mathfrak{B}(\exp^{tu})\mathfrak{B}(\exp^{sv}\operatorname{Ad}_{\mathfrak{B}(\exp^{sv})}\exp^{-tu}) \\
= \quad & \frac{d^2}{dtds}\Big|_{t,s=0} \mathfrak{B}(\exp^{tu}(\operatorname{Ad}_{\mathfrak{B}(\exp^{tu})}\exp^{sv})(\operatorname{Ad}_{\mathfrak{B}(\exp^{tu})\mathfrak{B}(\exp^{sv})}\exp^{-tu})) \\
= \quad & \mathfrak{B}_{*e}\left(\frac{d^2}{dtds}\Big|_{t,s=0}\operatorname{Ad}_{\mathfrak{B}(\exp^{tu})}\exp^{sv} + \frac{d^2}{dtds}\Big|_{t,s=0}\operatorname{Ad}_{\mathfrak{B}(\exp^{sv})}\exp^{-tu}\right. \\
& \qquad \left. + \frac{d^2}{dtds}\Big|_{t,s=0}\exp^{tu}\exp^{sv}\exp^{-tu}\right) \\
= \quad & B([B(u), v] - [B(v), u] + [u, v]).
\end{aligned}
$$



From Rota-Baxter operators to post-groups

Theorem (Bai-Guo-S.-Tang)

Let $\mathfrak{B} : G \longrightarrow G$ be a Rota-Baxter operator on a group $(G, \cdot)$. Define a binary product $\triangleright : G \times G \to G$ as following:

$$g \triangleright h = \mathrm{Ad}_{\mathfrak{B}(g)} h, \quad \forall g, h \in G.$$

Then $(G, \cdot, \triangleright)$ is a post-group.

From post-groups to Rota-Baxter operators

Proposition (Bai-Guo-S.-Tang)

Let $(G, \cdot, \triangleright)$ be a post-group. Then the identity map $\mathrm{Id} : G \to G$ is a relative Rota-Baxter operator on the subadjacent group $(G, \circ)$ with respect to the action $L^{\triangleright}$ on the group $(G, \cdot)$.

Skew-left braces

Definition (Rump)

A skew-left brace $(G, \circ, \cdot)$ consists of a group $(G, \cdot)$ and a group $(G, \circ)$ such that

$$a \circ (b \cdot c) = (a \circ b) \cdot a^{\cdot -1} \cdot (a \circ c), \quad \forall a, b, c \in G.$$


W. Rump, A decomposition theorem for square-free unitary solutions of the quantum Yang-Baxter equation. **Adv. Math.** 193 (2005), 40-55.


T. Gateva-Ivanova, Set-theoretic solutions of the Yang-Baxter equation, braces and symmetric groups. **Adv. Math.** 338 (2018), 649-701.


F. Cedó, A. Smoktunowicz and L. Vendramin, Skew left braces of nilpotent type. **Proc. Lond. Math. Soc.** 118 (2019), 1367-1392.

Proposition (Bai-Guo-S.-Tang)

Let $(G, \circ, \cdot)$ be a skew-left brace. Define a binary product $\triangleright : G \times G \to G$ by

$$a \triangleright b = a^{-1} \cdot (a \circ b), \quad \forall a, b \in G,$$

here a^{-1} is the inverse of a in $(G, \cdot)$. Then $(G, \cdot, \triangleright)$ is a post-group.

Skew-left braces

Proposition (Bai-Guo-S.-Tang)

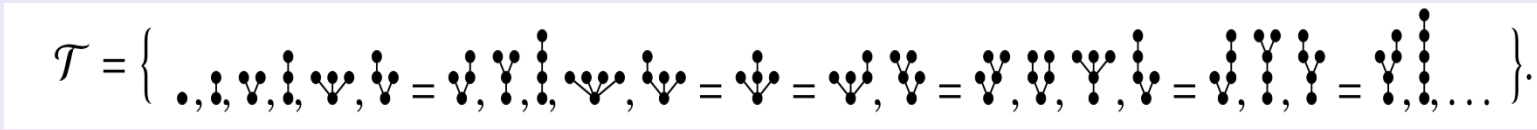
Let $(G, \cdot, \triangleright)$ be a post-group. Then $(G, \circ, \cdot)$ is a skew-left brace.

Theorem (Bai-Guo-S.-Tang)

The category of post-groups is isomorphic to the category of skew-left braces.

Butcher groups

Let $\mathcal{T}$ be the set of isomorphism classes of rooted trees:

$\mathcal{T} =$ 

We set $\mathcal{T}^+ = \mathcal{T} \cup \{\emptyset\}$ and denote by

$$\mathcal{B}_{\mathbb{R}} = \{a : \mathcal{T}^+ \to \mathbb{R} | a(\emptyset) = 1\}.$$

Theorem (Hairer-Wanner)

$(\mathcal{B}_{\mathbb{R}}, \circ)$ is a group, which is called **Butcher group**, *where*

$$(a \circ b)(\tau) = a(\tau) + \sum_{c \in AC(\tau)} a(P^c(\tau)) b(R^c(\tau)).$$

E. Hairer and G. Wanner, On the Butcher group and general multi-value methods, *Computing* 13 (1974), 1-15.

Butcher group

We define an abelian group structure on $\mathcal{B}_{\mathbb{R}}$ by

$$(a \cdot b)(\emptyset) = 1, \ (a \cdot b)(\omega) = a(\omega) + b(\omega), \ \forall \omega \in \mathcal{T},$$

Define the binary product $\triangleright : \mathcal{B}_{\mathbb{R}} \times \mathcal{B}_{\mathbb{R}} \to \mathcal{B}_{\mathbb{R}}$ by

$$\begin{aligned} (a \triangleright b)(\emptyset) &= 1, \\ (a \triangleright b)(\tau) &= \sum_{c \in AC(\tau)} a(P^c(\tau)) b(R^c(\tau)), \forall \omega \in \mathcal{T}. \end{aligned}$$

Theorem (Bai-Guo-S.-Tang)

With the above notations, $(\mathcal{B}_{\mathbb{R}}, \cdot, \triangleright)$ is a post-group, whose subadjacent group is exactly the Butcher group $(\mathcal{B}_{\mathbb{R}}, \circ)$.

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The Yang-Baxter equation

We show that a post-group gives rise to a braiding group, and thus lead to a set-theoretical solution of the Yang-Baxter equation.

Definition

Let X be a set. A set-theoretical solution to the **Yang-Baxter equation** on X is a bijective map $R : X \times X \to X \times X$ satisfying:

$$R_{12}R_{23}R_{12} = R_{23}R_{12}R_{23}.$$


V. G. Drinfel'd, On some unsolved problems in quantum group theory. Quantum groups (Leningrad, 1990), 1-8, Lecture Notes in Math., 1510, Springer, Berlin, 1992.


P. Etingof, T. Schedler and A. Soloviev, Set-theoretical solutions to the quantum Yang-Baxter equation. *Duke Math. J.* **100** (1999), 169-209.


J. Lu, M. Yan and Y. Zhu, On the set-theoretical Yang-Baxter equation. *Duke Math. J.* **104** (2000), 1-18.

Yang-Baxter equations

Let $(G, \cdot, \triangleright)$ be a post-group. Define $R_G : G \times G \to G \times G$ by

$$R_G(x, y) = (x \triangleright y, (x \triangleright y)^{\dagger} \circ x \circ y), \ \forall x, y \in G,$$

where $\circ$ is the subadjacent group structure.

Theorem (Bai-Guo-S.-Tang)

Let $(G, \cdot, \triangleright)$ be a post-group. Then $\big((G, \circ), R_G\big)$ is a braiding group, and R_G is a solution of the Yang-Baxter equation on the set G.

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   - Quiver-theoretical solutions of the Yang-Baxter equation

Definition

A **post-Lie algebroid** structure on a vector bundle $A \longrightarrow M$ is a triple that consists of a $C^\infty(M)$-linear Lie algebra structure $[\cdot,\cdot]_A$ on $\Gamma(A)$, a bilinear operation $\triangleright_A : \Gamma(A) \otimes \Gamma(A) \longrightarrow \Gamma(A)$ and a vector bundle morphism $a_A : A \longrightarrow TM$, called the **anchor**, such that $(\Gamma(A), [\cdot,\cdot]_A, \triangleright_A)$ is a post-Lie algebra, and for all $f \in C^\infty(M)$ and $u, v \in \Gamma(A)$, the following relations are satisfied:

(i) $u \triangleright_A (fv) = f(u \triangleright_A v) + a_A(u)(f)v$,

(ii) $(fu) \triangleright_A v = f(u \triangleright_A v)$.

H. Z. Munthe-Kaas and A. Lundervold, On post-Lie algebras, Lie-Butcher series and moving frames, *Found. Comput. Math.* **13** (2013), 583-613.

Example

Let $(\mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}})$ be a Lie algebra and M a manifold. Then the section space $\Gamma(M \times \mathfrak{g})$ of the trivial bundle $A = M \times \mathfrak{g}$ enjoys a $C^\infty(M)$-linear Lie algebra structure $[\cdot,\cdot]_{\mathfrak{g}}$ given by

$$[fu, gv]_{\mathfrak{g}} = fg[u,v]_{\mathfrak{g}}, \quad \forall f, g \in C^\infty(M),\ u, v \in \mathfrak{g}.$$

Let $\phi : \mathfrak{g} \to \mathfrak{X}(M)$ be an action of $\mathfrak{g}$ on M. Then one can define $a_A : M \times \mathfrak{g} \to TM$ by

$$a_A(m,u) = \phi(u)_m, \quad \forall m \in M, u \in \mathfrak{g}$$

and define $\triangleright_A : \Gamma(A) \times \Gamma(A) \longrightarrow \Gamma(A)$ by

$$(fu) \triangleright_A (gv) = f\phi(u)(g)v, \quad \forall f, g \in C^\infty(M),\ u, v \in \mathfrak{g}.$$

Then $(A = M \times \mathfrak{g}, [\cdot,\cdot]_{\mathfrak{g}}, \triangleright_A, a_A)$ is a post-Lie algebroid.

Definition 2.5. A **post-(Lie) groupoid** consists of the following data:

- a (Lie) group bundle $(\mathbb{G} \xrightarrow{\pi} M, \cdot)$, where $\cdot$ is the multiplication;
- a surjective map (submersion) $\Phi : \mathbb{G} \to M$ satisfying $\Phi(\iota_m) = m$, for all $m \in M$;
- a (smooth) map $\triangleright : \mathbb{G}_\Phi \times_\pi \mathbb{G} \to \mathbb{G}$ satisfying $\pi(\gamma) = \pi(\gamma \triangleright \delta)$ and the left multiplication

$$L_\gamma^{\triangleright} : \mathbb{G}_{\Phi(\gamma)} \to \mathbb{G}_{\pi(\gamma)}, \quad L_\gamma^{\triangleright}\delta = \gamma \triangleright \delta \text{ for all } \delta \in \mathbb{G}_{\Phi(\gamma)},$$

is invertible for any $\gamma \in \mathbb{G}$, where

$$\mathbb{G}_\Phi \times_\pi \mathbb{G} = \{(\gamma, \delta) \in \mathbb{G} \times \mathbb{G} \text{ such that } \Phi(\gamma) = \pi(\delta)\},$$

such that for all $(\gamma_1, \gamma_2, \gamma_3) \in \mathbb{G}_\Phi \times_\pi \mathbb{G}_\Phi \times_\pi \mathbb{G}$, and $\gamma_2' \in \mathbb{G}$ satisfying $\pi(\gamma_2) = \pi(\gamma_2')$, the following axioms hold:

(i) $\Phi(\gamma_2) = \Phi\Big(\gamma_1 \cdot (\gamma_1 \triangleright \gamma_2)\Big)$,
(ii) $\gamma_1 \triangleright (\gamma_2 \cdot \gamma_2') = (\gamma_1 \triangleright \gamma_2) \cdot (\gamma_1 \triangleright \gamma_2')$,
(iii) $\gamma_1 \triangleright (\gamma_2 \triangleright \gamma_3) = \Big(\gamma_1 \cdot (\gamma_1 \triangleright \gamma_2)\Big) \triangleright \gamma_3$.

We will denote a post-(Lie) groupoid by $(\mathbb{G} \xrightarrow{\pi} M, \cdot, \Phi, \triangleright)$.

Example

Let G be a group and M a set. Then $\mathbb{G} = M \times G \xrightarrow{\pi} M$ is a trivial group bundle, where π is the projection to M. Let $\Phi : M \times G \to M$ be an action of G on M. In this case,

$$\mathbb{G}_\Phi \times_\pi \mathbb{G} = \{((m,g),(n,h)) \in \mathbb{G} \times \mathbb{G} \text{ such that } n = \Phi(m,g)\}.$$

Define $\triangleright : \mathbb{G}_\Phi \times_\pi \mathbb{G} \to \mathbb{G}$ by

$$(m,g) \triangleright (n,h) = (m,h).$$

Then $(M \times G \xrightarrow{\pi} M, \cdot, \Phi, \triangleright)$ is a post-groupoid.

Theorem (S. -Tang-Zhu)

Let $(\mathbb{G} \xrightarrow{\pi} M, \cdot, \Phi, \triangleright)$ *be a post-groupoid. Then* $(\mathbb{G} \underset{\beta=\Phi}{\overset{\alpha=\pi}{\rightrightarrows}} M, \star, \iota, \mathrm{inv}_{\triangleright})$ *is a groupoid, called the* **sub-adjacent groupoid**, *where the groupoid multiplication* $\star : \mathbb{G}_\Phi \times_\pi \mathbb{G} \to \mathbb{G}$ *is defined by*

$$\gamma_1 \star \gamma_2 = \gamma_1 \cdot (\gamma_1 \triangleright \gamma_2),$$

and the new inverse map $\mathrm{inv}_{\triangleright}$ *is given by*

$$\mathrm{inv}_{\triangleright}(\gamma) = (L_\gamma^{\triangleright})^{-1}\mathrm{inv}(\gamma).$$

Moreover, $\triangleright$ *gives rise to an action of the sub-adjacent groupoid* $(\mathbb{G} \underset{\beta=\Phi}{\overset{\alpha=\pi}{\rightrightarrows}} M, \star, \iota, \mathrm{inv}_{\triangleright})$ *on the group bundle* $\mathbb{G} \xrightarrow{\pi} M$.

Denote the Lie algebroid of the subadjacent Lie groupoid $(\mathbb{G} \underset{\beta=\Phi}{\overset{\alpha=\pi}{\rightrightarrows}} M, \star, \iota, \mathrm{inv}_{\triangleright})$ by $(A \longrightarrow M, \lceil \cdot, \cdot \rceil, \Phi_*)$. The Lie groupoid homomorphism $L^{\triangleright}$ induces a Lie algebroid homomorphism $L^{\triangleright}_*$ from $(A \longrightarrow M, \lceil \cdot, \cdot \rceil, \Phi_*)$ to $\mathrm{Der}(A)$. In particular, there holds:

$$\Phi_*(u) = \mathfrak{a}(L^{\triangleright}_*(u)), \quad \forall u \in \Gamma(A).$$

Define $\triangleright_A : \Gamma(A) \otimes \Gamma(A) \to \Gamma(A)$ by

$$u \triangleright_A v = L^{\triangleright}_*(u)v, \quad \forall u, v \in \Gamma(A).$$

Theorem (S. -Tang-Zhu)

Let $(\mathbb{G} \xrightarrow{\pi} M, \cdot, \Phi, \triangleright)$ be a post-Lie groupoid. Then $(A \longrightarrow M, [\cdot, \cdot]_A, \Phi_*, \triangleright_A)$ is a post-Lie algebroid, where $(A \longrightarrow M, [\cdot, \cdot]_A)$ is the Lie algebra bundle associated to the Lie group bundle $\mathbb{G} \xrightarrow{\pi} M$.

Definition

A **quiver** over M is a set $\mathbb{A}$ equipped with two maps $\alpha, \beta : \mathbb{A} \to M$, called the source map and the target map.

Definition

Let $(\mathbb{A} \underset{\beta}{\overset{\alpha}{\rightrightarrows}} M)$ be a quiver. A quiver-theoretical solution of the **Yang-Baxter equation** on the quiver $\mathbb{A}$ is an isomorphism R of quivers from $\mathbb{A}_\beta \times_\alpha \mathbb{A}$ to $\mathbb{A}_\beta \times_\alpha \mathbb{A}$ satisfying:

$$R_{12}R_{23}R_{12} = R_{23}R_{12}R_{23}.$$


N. Andruskiewitsch, On the quiver-theoretical quantum Yang-Baxter equation. *Selecta Math. (N.S.)* **11** (2005), 203-246.

Let $(\mathbb{G} \xrightarrow{\pi} M, \Phi, \triangleright)$ be a post-groupoid. Define $R_{\mathbb{G}} : \mathbb{G}_\Phi \times_\pi \mathbb{G} \to \mathbb{G}_\Phi \times_\pi \mathbb{G}$ by

$$R_{\mathbb{G}}(\gamma, \delta) = \big(\gamma \triangleright \delta, \mathrm{inv}_\triangleright(\gamma \triangleright \delta) \star \gamma \star \delta\big), \quad \forall (\gamma, \delta) \in \mathbb{G}_\Phi \times_\pi \mathbb{G}.$$

Theorem (S. -Tang-Zhu)

Let $(\mathbb{G} \xrightarrow{\pi} M, \cdot, \Phi, \triangleright)$ be a post-groupoid. Then $R_{\mathbb{G}}$ is a quiver-theoretical solution of the Yang-Baxter equation on the quiver $\mathbb{G} \underset{\beta=\Phi}{\overset{\alpha=\pi}{\rightrightarrows}} M$.

Thanks for your attention!